

A PSO-Based Fuzzy-Controlled Searching for the Optimal Charge Pattern of Li-ion Batteries

Shun-Chung Wang and Yi-Hua Liu, *Member, IEEE*

Abstract—This paper proposes a searching strategy based on particle swarm optimization (PSO), combined with a fuzzy-deduced fitness evaluator (FDPE), to find the optimal multistage charging pattern that deliver the most discharged capacity within the shortest charging time. The objective function of the optimization problem is to maximize the cost-effectiveness for the applied charging pattern based on the charging time (CT) and normalized discharged capacity (NDC). Therefore, this paper proposes an FDPE to combine CT and NDC into a unified cost function to properly evaluate the multiple performance characteristics index in the charge problem. The experimental results show that the obtained pattern is capable of charging the batteries to over 88% capacity within 51 min. Compared with the conventional constant current–constant voltage (CC-CV) method, the charging time, obtained life cycle and the charging efficiency of the lithium-ion (Li-ion) battery for the devised approach are improved by approximately 56.8%, 21% and 0.4%, respectively. The presented charging approach is suitable for the increasingly applications in which the batteries are “sealed” inside the products to extend the lifespan.

Index Terms—Constant current–constant voltage (CC-CV), fuzzy control, Lithium-ion battery, multistage charge pattern, particle swarm optimization (PSO).

I. INTRODUCTION

RECHARGEABLE batteries play a substantial role in energy storage solutions for modern-day technologies such as hand-held consumer electronics, aerospace, electric vehicles and energy backup for renewable energy systems, etc. Among the commonly used rechargeable batteries (lead-acid, nickel-based, and lithium-ion (Li-ion) chemistries), the Li-ion batteries are a popular choice for contemporary electronic appliances [1]–[6]. This is because of attractive characteristics such as being environmentally friendly and demonstrating high energy and power density, high open-circuit voltage (OCV), low self-discharge rate, low maintenance, and immunization of memory effect. Li-ion batteries, however, are highly sensitive

to deep discharge or overcharge because of the improper charge or discharge C rate (C represents the battery rated capacity). Since the available functionality and lifetime of the Li-ion batteries are strongly dependent on their electrochemistry, operating temperature and depth of discharge level. Hence, a well-designed charger must be equipped with optimal charging policies and patterns to maximize discharged capacity, shorten charging time, and lengthen the cycle life of the battery [7]–[11].

Rapid charging has always been an essential design consideration for the battery-powered devices. Therefore, determining how to make rapid charging achievable, yet still maintain high efficacy, has become a focus of charger development [12]–[20]. A commonly adopted constant current–constant voltage (CC-CV) charging profile for commercial Li-ion batteries can be divided into three distinct phases as illustrated in Fig. 1 [1], [9]. The trickle charge is conducted only if the battery is deeply discharged. During trickle charge, the battery is charged with a low current. It switches to CC mode when the voltage rises to the threshold value (typically 3.0 V). In CC mode, a higher constant current is utilized to charge the battery till a pre-defined voltage level (typically 4.2 V) is reached, and then a CV mode is applied until the end-of-charge condition is met. To prevent batteries from overcharge in CV mode, the charging current should be reduced tardily. The CV stage is over until the charge current drops below a minimum threshold (typically 0.01C). However, the CC-CV method is unsuitable for rapid charging because the CV charging phase substantially prolongs the charging time and also degrades the charging efficiency.

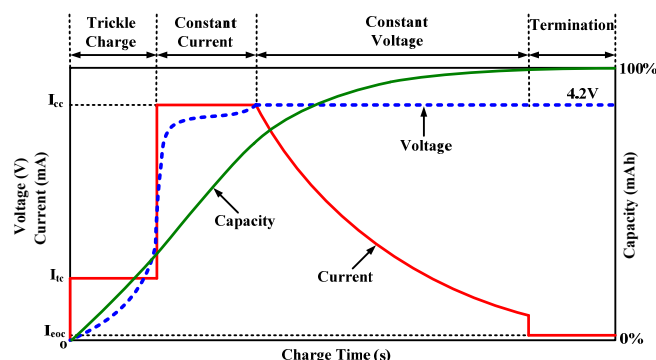


Fig. 1. A typical CC-CV charging cycle for Li-ion batteries.

To address this problem, several studies have investigated charging techniques and topologies. A phase-shift full-bridge converter is proposed in [21] as a Li-ion battery charger that offers higher output capability and conversion efficiency. In [22], a smooth control circuit with a built-in resistance

Manuscript received March 25, 2014; revised July 10, 2014 and August 23, 2014; accepted September 16, 2014.

Copyright © 2014 IEEE. Personal use of this material is permitted. However, permission to use this material for any other purposes must be obtained from the IEEE by sending a request to pus-permissions@ieee.org

This work was supported by the Ministry of Science and Technology, Taiwan, under Grant MOST 103-2221-E-262 -012.

Shun-Chung Wang is with the Department of Electrical Engineering, Lunghwa University of Science and Technology, Taoyuan 33306, Taiwan. (e-mail: wangsc@mail.lhu.edu.tw)

Yi-Hua Liu is with the Department of Electrical Engineering, National Taiwan University of Science and Technology, Taipei, 106, Taiwan. (e-mail: yhliu@mail.ntust.edu.tw)

compensator method is presented to determine the transition time from the CC to the CV stage and expedite the charging process. Ramadass et al. developed a capacity fade model of Li-ion cells based on first principles to simulate and analyze the cycle life and the discharge characteristics. The effects of parameters such as end of charge voltage, depth of discharge, and film resistance etc. on the capacity fade and battery performance during CC-CV charging were studied qualitatively [23]. In [24]-[27], Chen et al. addressed a series of improvements in charging strategies, waveforms, or circuit topologies to satisfy the requirement of rapid charging. These methods and schemes improved charging efficiency and speed. In [28]-[32], the advantages of the multistage CC-charging methods have been confirmed. The benefits of multistage CC-charging scheme include less chemical reaction stress and shorter charging time. Recently, intelligent control algorithms, such as fuzzy control, neural network, grey-prediction control, and genetic approaches [33]-[38], have been integrated into microcontrollers to improve the accuracy of the parameter prediction and operating performance of the charger. In spite of the progress in charger design, accurate operating parameters and mathematical models of the battery are necessary for these algorithms as the analysis and computation are processed. However, a precise model of a Li-ion battery is difficult to obtain because of its complicated electrochemical properties. Therefore, improvement in charging modes alone could not promise to obtain optimal charging profiles and significantly boost the charging speed and efficacy. Furthermore, battery cycle life issues are seldom discussed in these researches.

Searching for the optimal charging pattern for a rechargeable battery can be regarded as a combinatorial optimization problem. Such a problem is difficult to resolve using traditional methods. One possible way to obtain the most favorable pattern is to attempt every combination of the charging current value. However, this way of exhaustive search is time-consuming and not economical for engineers. Global optimization techniques, such as genetic algorithm (GA) [37], non-dominated sorting genetic algorithm II (NSGA-II) [38], and the ant colony system (ACS) [39], have been successfully applied to attain improved charging performance, including charging time, battery longevity, and energy cost. The Taguchi-based approach cooperated with consecutive orthogonal array techniques have also been proven to be feasible and effective in determining the optimal charging pattern of the five-step CC charging mechanism [40, 41].

This paper introduces a global optimization technique based on the PSO algorithm with fuzzy control to solve the rapid charging problem. The cost-effectiveness of the charging pattern is formulated by two dependent parameters, i.e. charging time (CT) and normalized discharged capacity (NDC). Allocated proportions of the weightings regarding the CT and NDC within the cost-effectiveness function are derived according to fuzzy logic inference for obtaining maximal charging benefit. The main contributions of this paper include that a simple and easier-to-implement optimization method (PSO) is applied for optimal charging pattern searching and,

the presented charging approach is suitable to the recent applications in which the batteries are “sealed” inside the products (such as Smartphone or some Tablet) to prolong the operating lifespan because the user cannot replace an aged battery on their own. The rest of this paper is organized as follows. Section II introduces the basic philosophy of the PSO algorithm. Fundamentals of the proposed PSO-based fuzzy-controlled searching method, including the principle of the five-stage CC charging pattern, problem formulation, fitness value evaluation and the searching procedures are described in Section III in detail. Section IV illustrates the test system configuration, parameter setting and operating process. Section V shows the experimental results and the comparisons with the CC-CV method. Finally, Section VI concludes this paper.

II. PHILOSOPHY OF PSO ALGORITHM

The PSO algorithm is developed by Eberhart and Kennedy first in 1995. It is a powerful tool for dealing with global optimization problems with single- or multi-objective functions containing several local optimum values [42]. The most remarkable property of the PSO approach is its simplicity and effectiveness because only two model equations are required. Several variants of the PSO algorithm are proposed for particular applications, yet the most popular is still the global formulation (G_{best} model) devised by Eberhart and Shi [43].

Compositions dominating the flying experience of each particle are shown in Fig. 2. In a problem with n-dimension search space, the movement of each particle is related to the velocity and position vectors. The position and velocity vectors can be represented as $\vec{x} = [x_1, x_2, \dots, x_n]$ and $\vec{v} = [v_1, v_2, \dots, v_n]$, respectively. Every particle will share information with the rest of the swarm in a certain communication way during their exploration and exploitation. These interaction aims at optimizing their searching experience, which contains individual cognitive part and global social part. The particle with the best solution shares its position (G_{best}) with the other members. Then, each particle updates its flying direction based on its own best local exploratory experience ($P_{best,i}$) and the best global one (G_{best}) according to the following equations:

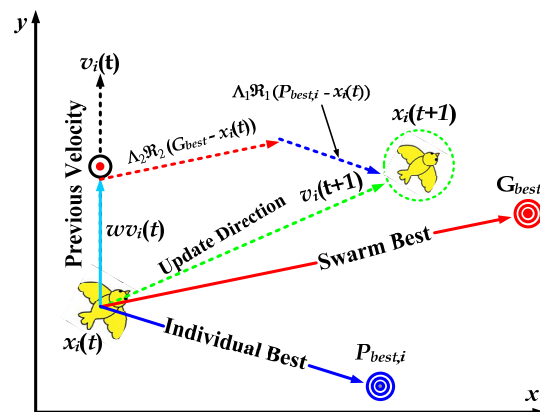


Fig. 2. Concept of update in a searching particle by PSO.

$$v_i^n(k+1) = \underbrace{w(k)v_i^n(k)}_{\text{previous velocity}} + \underbrace{\Lambda_1 \mathfrak{R}_1 (P_{best,i}^n - x_i^n(k))}_{\text{cognitive component}} + \underbrace{\Lambda_2 \mathfrak{R}_2 (G_{best}^n - x_i^n(k))}_{\text{social component}} \quad (1)$$

$$x_i^n(k+1) = x_i^n(k) + v_i^n(k+1) \quad (2)$$

Where $x_i^n(k+1)$ and $x_i^n(k)$ represent the current and previous positions in the n th dimension for particle i , respectively; $v_i^n(k+1)$ and $v_i^n(k)$ denote the current and previous velocities in the n th dimension, respectively. The $w(k)$ is the inertia weight. The Λ_1 and Λ_2 are the cognitive and social learning factors, respectively. The $\mathfrak{R}_1$ and $\mathfrak{R}_2$ are two random numbers in the range $[0, 1]$, respectively. G_{best}^n is the location with the best global fitness value. $P_{best,i}^n$ is the location at which the particle i previously had the best fitness value.

III. PSO-BASED FUZZY-CONTROLLED FAST CHARGING PATTERN SEARCH

A. Five-Stage Constant-Current Charging

This paper chooses the five-stage CC (5SCC) charging approach for performing rapid charge of Li-ion batteries because it has been confirmed to fulfill the rapid charge criteria (i.e. acquire more capacity in shorter charging time) and particularly mitigate the chemical reaction stress and temperature rise of the batteries under charge [28]-[31]. Ref. [32] examines and tests the effect of different numbers of current levels (stages) on the optimal charging time and charging efficiency. According to the experimental results, the difference between applying different number of charging current levels is almost negligible while the number of charging current levels is more than five. Therefore, a gradual decreasing five-stage charging levels are chosen in this study. Fig. 3 illustrates the basic concept of the multistage CC charging method. The total charging period, as shown in Fig. 3, is divided into five stages. The charging current in each stage is preset to a default value. During the charging phase, the battery voltage will increase gradually. The charging process will be switched to the next stage and a new set of charge current will be applied accordingly as soon as the voltage arrives at the default limit voltage (V_{lim} , typically 4.2V). This process will continue until the charging stage number reaches 5. The gradual reduction mechanism of the charging current makes the multistage CC method easier for balancing the chemical reaction and subduing the temperature rise of the battery, resulting in shorter charging time and more running cycles.

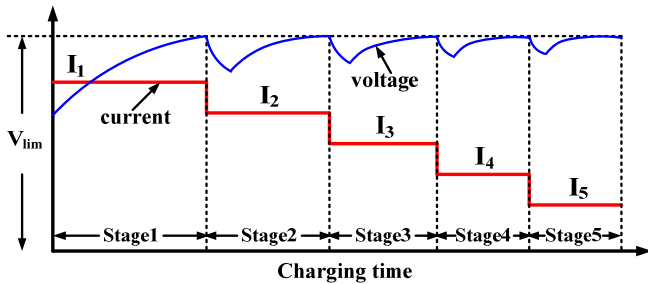


Fig. 3. Basic structure of the multistage CC charging method.

B. Problem Formulation

This study determines the optimal five-stage charging pattern to meet the rapid charging criteria and obtain the best charging benefit. Therefore, the objective function of this problem can be formulated as

$$\text{Maximize } f_{ce}^i(T_{ch}^i(\bar{I}_j^i), C_{dis-norm}^i(I_{dis}^i)), \quad \bar{I}_j^i \in \mathbb{S}$$

$$\text{Where } \bar{I}_j^i = [I_1^i, I_2^i, I_3^i, I_4^i, I_5^i]$$

$$\text{Subject to } 0 \min \leq T_{ch}^i(\bar{I}_j^i) \leq 90 \min \quad (3)$$

$$80\% \leq C_{dis-norm}^i(I_{dis}^i) \leq 100\%$$

$$I_j^i \geq I_m^i \quad \text{if } j < m, \quad j, m = 1, 2, \dots, 5$$

In (3), $\mathbb{S}$ denotes the set of all feasible charging patterns, and i is the battery number. The objective function is to maximize the cost-effectiveness function f_{ce}^i which is composed of two performance index, i.e. charging time (CT) T_{ch}^i and normalized discharged capacity (NDC) $C_{dis-norm}^i$. The NDC is defined as the percentage of the discharged capacity (with C/10 discharge current) obtained by the applied charging pattern to initial rated capacity, which can be expressed by

$$C_{dis-norm}^i = \frac{C_{dis-5-stage}^i}{C_{rated}^i} \quad (4)$$

where $C_{dis-5-stage}^i$ and C_{rated}^i are the discharged capacities obtained from the 5SCC charging and the initial rated capacity of battery i , respectively.

A detailed description of how to formulate the objective function $f_{ce}^i(\cdot)$ is provided in Section III.C. In (3), any constraints related to the optimization problem should be incorporated into the objective function as penalty factors. The constraints on the CT and NDC, in this paper, are specified to the range $[30, 90]$ min and $[80\%, 100\%]$, respectively, to meet the common considerations of the rapid charging criterion. Also, the charging current of the present stage should not be greater than that of the preceding stage to alleviate the chemical reaction stress and increase the charge capacity.

C. Fuzzy-Deduced Fitness Evaluator (FD FE)

Fitness value in the PSO-based optimization problem is used to evaluate the performance of the applied charging pattern. The obtained best charging pattern should make the batteries deliver the most discharged capacity within the shortest charging time. Hence, the fitness function $F^i(\cdot)$ for the battery i can be expressed as

$$F^i(T_{ch}^i, C_{dis-norm}^i) = f_{\alpha}^i(T_{ch}^i(\bar{I}_j^i)) + f_{\beta}^i(C_{dis-norm}^i(I_{dis}^i)) \quad (5)$$

Where the factors $f_{\alpha}^i(\cdot)$ and $f_{\beta}^i(\cdot)$ are the weightings of the CT and NDC respectively, which are added to highlight their contributions to the cost function.

However, according to the electrochemical characteristics of and the field-tested experience with the battery, the charging time and discharged capacity are strongly dependent on the

charging and discharging current levels, and the relationship between the charging time and charged capacity are typically opposite to each other. Therefore, exactly how to determine the weighting factor dispatch to acquire the best fitness for each updating process is the primary outcome of the proposed approach. A constant 50% dispatch for $f_{\alpha}^i(\cdot)$ and $f_{\beta}^i(\cdot)$ cannot always provide best fitness. In contrast, the fuzzy logic controller approximates specialist intuitiveness, which makes it attractive and easy to incorporate heuristic rules that reflect the experience of human experts into the FDFE. Accordingly, this paper proposes an FDFE. The scheme of the proposed FDFE for the fitness value assessment is shown in Fig. 4. As shown in Fig. 4, the CT and NDC are used as the input variables, and the output variable is the evaluative value of the fitness function, F_N^i . In the fuzzy controller, the Mamdani-type minimum inferential method cooperating with the center-of-area (COA) defuzzification procedure is utilized to produce the crisp evaluator output. The COA computes the center of gravity of the reasoned results to obtain the crisp output and the calculation is indicated by

$$y = \frac{\sum_{h=1}^m W_h B_h}{\sum_{h=1}^m W_h} \quad (6)$$

Where W_h is the inferential result of rule h , B_h is the corresponding output of rule h , m is the rule number, and y is the crisp output.

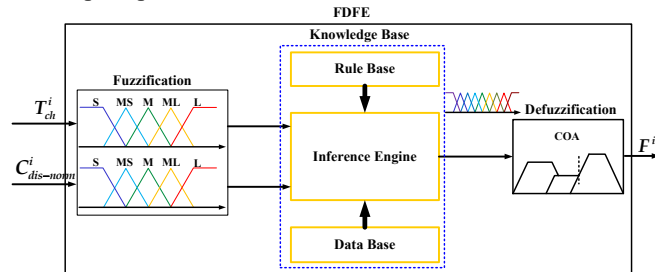


Fig. 4. Scheme of the proposed FDFE.

The membership functions (MFs) corresponding to the NDC, CT, and fitness value are illustrated in Fig. 5(a)-(c), respectively. Here in all three MFs, symmetric triangles with equal base and fifty-percent overlap with adjacent MFs are chosen. The universe of discourse (UOD) in the MFs of CT and NDC are defined on the domain [30 min, 90 min] and [80%, 100%], respectively. The UOD of the output MF is defined on the normalized domain [0, 1]. In Fig. 5, the linguistic values VS, SS, MS, M, ML, LL, and VL represent very small, small, medium small, medium, medium large, large, and very large, respectively. From Fig. 5, each of the input variables NDC and CT is mapped into 5 different linguistic values. Therefore, the proposed fuzzy rule table will consist of 25 different rules. The complete rule bases for the proposed FDFE are illustrated in Table I. As shown in Table I, the dispatch rules are deduced to obtain the optimal cost effectiveness function based on the perception that the larger discharged capacity and shorter charging time produce the superior fitness value and result in a

higher charging cost effectiveness.

Consequently, the weighting distribution is driven by a set of control rules rather than constant proportional gain. The FDFE emulates expert experience to achieve appropriate weighting dispatch, making it easy to incorporate heuristic rules that reflect specialist experience into the calculating processes.

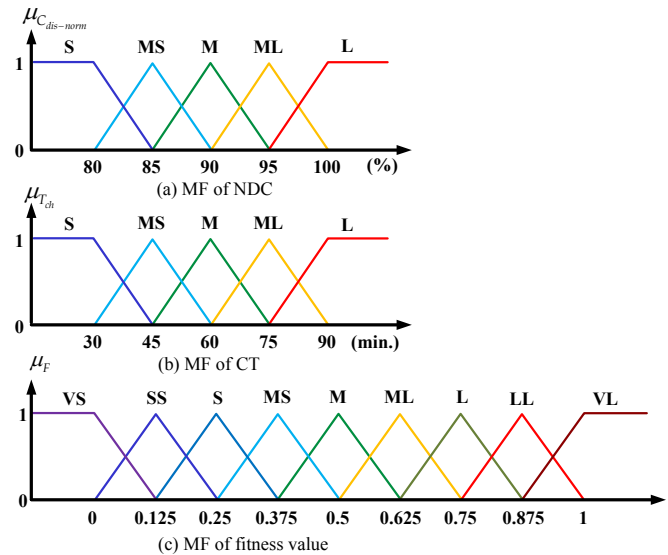


Fig. 5. MFs of the input & output variables.

TABLE I RULE BASED DERIVATION FOR FITNESS EVALUATION						
T_{ch}^i	$C_{dis-norm}^i$					
	S	MS	M	ML	L	
S	M	ML	L	LL	VL	Rule1
MS	MS	M	ML	L	LL	Rule6
M	S	MS	M	ML	L	Rule11
ML	SS	S	MS	M	ML	Rule16
L	VS	S	S	MS	M	Rule21
						Rule2
						Rule7
						Rule12
						Rule17
						Rule22
						Rule3
						Rule8
						Rule13
						Rule18
						Rule23
						Rule4
						Rule9
						Rule14
						Rule19
						Rule24
						Rule5
						Rule10
						Rule15
						Rule20
						Rule25

D. PSO-Based Searching Procedure

Shown in Fig. 6 is the flowchart of the PSO-based fuzzy-controlled searching algorithm applied in this paper. As shown in Fig. 6, the main operating procedures are described as follows:

- Step1.) Battery screening: The specifications of the used Li-ion batteries are nominal voltage 3.6 V and rated capacity 2200 mAh. To enhance the reliability and validity of the experimental result, internal impedances of 55 new, commercially available Li-ion batteries were screened and, among them, 16 batteries with approximate impedances and similar electrochemical characteristics were selected for performing experiments.
- Step2.) Initial charging pattern generation and PSO-parameter design: To ensure broad coverage of the search space, the particles should be initialized so that they are

distributed as evenly as possible throughout the space. In this paper, 16 particles are applied to search for the optimal charging pattern. The initial charging patterns of each battery for stage 1 are generated randomly. In addition, the parameters formulated in (1) must be accurately established. The effect of the parameter setting on the algorithm is discussed in Section IV.

Step3.) Charge and discharge batteries: The charging process was conducted using a programmable battery tester (WonATech WBCS3000). When the charging processes are completed with the applied charging patterns, the discharging processes are executed using the same equipment, after an appropriate period of rest to balance the electrochemical reaction. Because a large discharging current tends to degrade the discharging efficiency, a small discharging current of 0.1C is employed to acquire a more accurate discharged capacity.

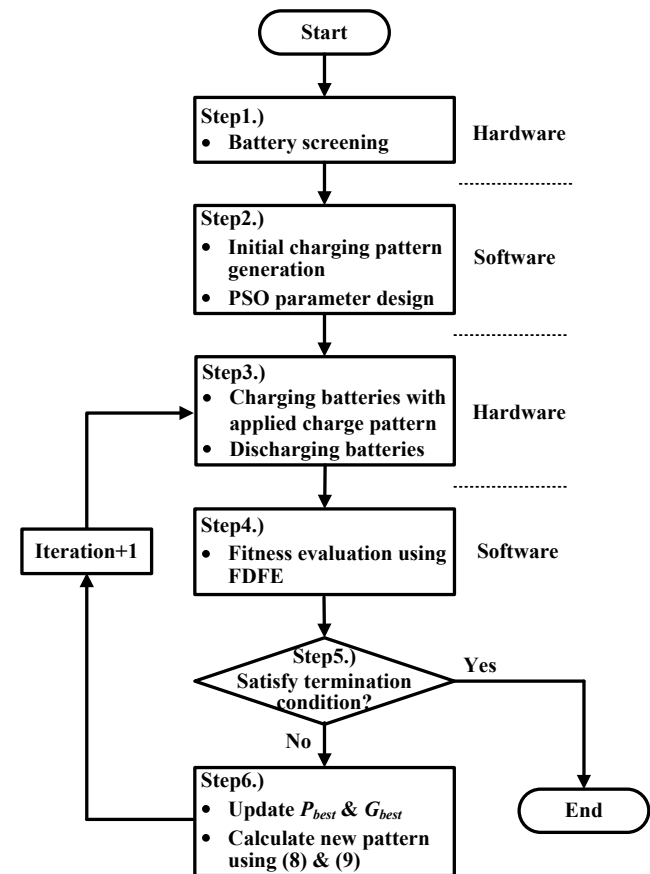


Fig. 6. Flowchart of the PSO-based algorithm.

Step4.) Fitness evaluation: The NDC and CT obtained in Step 3 were inputted into the FDFE to determine the fitness value of each particle in the swarm.

Step5.) Convergence determination: According to the electrochemical properties of the battery and the characteristics of the proposed PSO-based searching technique, two convergence criteria exist: a) the difference of the average fitness value (AFV) between

the before-and-after iteration is less than 0.005; b) the standard deviation (STDEV) of the FDFE output is less than 0.01. If criterion a) is satisfied, it indicates that the improvement is negligible. The STDEV in criterion b) is an uncertainty measurement that is used to estimate the deviation between the observations and their average value, and can be denoted as

$$STDEV = \sqrt{\frac{\sum_{i=1}^n s_i - \bar{s}}{n-1}} \quad (7),$$

where s_i is the sample, $\bar{s}$ is the average of the samples, and n is the sample number.

Step6.) Data update and new charging pattern generation: If the termination condition is not achieved, then the data of P_{best} and G_{best} must be updated to generate new charging patterns for the next experiment. According to (1) and (2), the update rules of the incremental current and charging current can be calculated as

$$\Delta I_i(k+1) = w(k)\Delta I_i(k) + \Lambda_1 \mathcal{R}_1(P_{best,i} - I_i(k)) + \Lambda_2 \mathcal{R}_2(G_{best} - I_i(k)) \quad (8)$$

$$I_i(k+1) = I_i(k) + \Delta I_i(k+1) \quad (9)$$

where the index k represents the iteration number, $\Delta I_i(k)$ and $\Delta I_i(k+1)$ are the incremental current for the i th battery at present and next iteration, respectively. $I_i(k)$ and $I_i(k+1)$ are the i th charging current for the present and next iteration, respectively.

IV. SETUP OF TEST SYSTEM AND SETTING OF ALGORITHM PARAMETER

A. Experimental System Architecture

Fig. 7 shows the configuration of the experimental system. The right-hand side of the dotted line is the software part, where the proposed PSO-based fuzzy-controlled searching algorithm is implemented using MATLAB on a Pentium PC. The left-hand side of the dotted line is the programmable battery tester. The charging/discharging equipment used in this paper is WBCS3000 (5 A, 5 V, 8 channel) from WonATech corporation. Referring to Fig. 7, the proposed PSO-based fuzzy-controlled searching algorithm is executed in the software part. Charging patterns are then programmed into this battery tester which controls and monitors current to a maximum resolution of 1 mA. Test data are sampled in every 1-s time interval. After the charging process is finished, the battery tester discharges the battery using a small discharging current (0.1C) to gauge the discharged capacity. The charging and discharging data retrieved by the battery tester are then transferred back to the PC in which the PSO-based fuzzy-controlled searching algorithm is implemented. After getting these data, the proposed algorithm calculates a new set of charging pattern using (8) and (9). The whole process will continue until the specified convergence conditions are met.

Sixteen commercially available Li-ion batteries with the same brand and specification (3.6V, 2200 mAh) from Samsung Corp. are utilized in these experiments. All the batteries possess

similar electrochemical characteristics and have passed screening tests with acceptable open circuit voltage (OCV), impedance, and rated capacity values. The batteries selected for test have the internal impedance ranges between 59.6 and 62.8 mΩ, the rated capacity ranges between 2114.3 and 2190.0 mAh, and the OCV greater than 3.63V. In addition, the initial rated capacity of each battery is recorded during the screening test. This value is important because the NDC in the fitness function should be evaluated as the capacity obtained from the C/10 discharging process divided by the initial rated capacity to eliminate the influence of the inherent capacity difference of batteries.

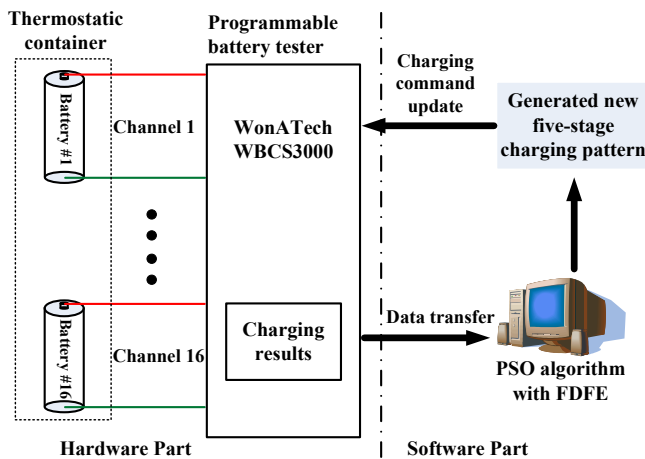


Fig. 7. Experimental system configuration.

On the other hand, the battery temperature rise comes from the drastic electrochemical reaction induced by the high charging or discharging current and, it leads to battery aging, capacity degeneration and life cycle shortening [23]. To decrease the influence of this aging factor on battery cycle life, a thermostatic container is employed. All the 16 batteries under test are put into the thermostatic container as shown in Fig. 8, and the operating temperature is kept constant as 25 °C. When the aging factor of the temperature rise has been excluded, the relationship between the different charging methods (or patterns) and the charge time, charge efficiency as well as lifetime of the battery can then be studied.

B. Parameter Setting of PSO Algorithm

Regarding the population sizes, according to (8), increasing the number of particles generally decreased the number of required iterations; however, the number of particles should also be proportional to the channel number required since all particles should be charged and discharged by the battery tester. The number of particles is chosen as 16 to ensure good convergence and accuracy as well as to take the full advantage of the battery tester channels. By gradually decreasing the inertia weight $w(k)$, the swarm has better tendency to eventually constrict itself down to the area involving the best fitness. Hence, the convergence characteristics of the PSO can be improved. The $w(k)$ with a linearly decreasing function of the current iteration index k was employed and can be denoted by



Fig. 8. Photograph of the battery test system.

$$w(k) = w_{max} - \frac{k}{k_{max}}(w_{max} - w_{min}) \quad (10)$$

Where k_{max} is the maximum number of the allowable iterations, the w_{max} and w_{min} are the maximum and minimum values of the inertia weight, respectively. The constants Λ_1 and Λ_2 are the cognitive and social learning factors, respectively. Choosing $\Lambda_1 > \Lambda_2$ would bias searching towards the direction of local optimum. In contrary, searching towards the direction of global optimum would be favored if $\Lambda_1 < \Lambda_2$. The $\mathfrak{R}_1$ and $\mathfrak{R}_2$ are two random numbers in the range [0, 1], which are added in the model to introduce stochastic nature to the particles' movement and provide the capability for particles to escape from the local optimum. In this paper, the parameter settings in the PSO model equations are: w_{min} is 0.1; w_{max} is 1.0; k_{max} is 20; Λ_1 and Λ_2 are equal to 2; $\mathfrak{R}_1$ and $\mathfrak{R}_2$ are random numbers between 0 and 1.

V. EXPERIMENTAL RESULTS

A. The First Experiment

The initial charging patterns for the first experiment were generated randomly. The yielded initial charging patterns for the first experiment are tabulated in Table II. Then the charging and discharging processes for the selected 16 charging patterns were carried out. Table II also shows the test results of the first experiment. The NDC and CT values for each battery were recorded. The acquired NDC and CT of the first charging pattern were then inputted into the FDFE to compute the fitness value (fuzzy output) for each battery. The result also shows that battery number 10 has the best fuzzy output. According to the evolutionary philosophy of PSO, the battery number 10 is chosen as the leading particle in the first exploration. The five-stage charging current (I_1 - I_5) levels of the other batteries are updated and modified toward those of battery number 10, based on its own optimal local exploratory experience.

B. The Second Experiment

To continue doing the second experiment, the new charging patterns of all batteries for the next iteration must first be determined. The incremental current $\Delta I_i(k+1)$ and new charging current $I_i(k+1)$ are determined using (8) and (9). The generated new charging patterns for the second experiment are shown in Table III. The test result and, the calculated fitness value of each battery for the first and second iterations are also inclusive for comparison purpose. From Table III, the data fields marked with green indicate that the fitness value obtained up to now is the local optimum. Because the STDEV and the difference of the AFV between the before-and-after iteration do not satisfy the convergence criteria as specified in the earlier statements, the search procedure should be continued until the convergence criteria are met. Hence, the third through the sixth experiments were conducted similarly.

TABLE II
TEST RESULTS OF THE FIRST CHARGE/DISCHARGE

The test result of the 1 st experiment (3.6V 2200mAh)												
Battery no.	Rate capacity (mAh)	I ₁	I ₂	I ₃	I ₄	I ₅	NDC (%)	CT (min)	Battery no.	Fuzzy Output1	Pbest	Gbest
B1	2286.95	1.32	1.23	0.96	0.52	0.43	0.8757	49.96	B1	0.516	0.516	
B2	2285.20	1.41	0.97	0.72	0.68	0.24	0.9252	64.71	B2	0.517	0.517	
B3	2266.90	0.89	0.74	0.53	0.41	0.12	0.9650	101.60	B3	0.417	0.417	
B4	2275.45	1.08	1.02	0.53	0.49	0.37	0.9019	60.40	B4	0.502	0.502	
B5	2278.25	1.17	1.02	0.91	0.56	0.45	0.8838	52.55	B5	0.515	0.515	
B6	2261.00	1.22	1.19	0.65	0.41	0.31	0.9189	59.55	B6	0.556	0.556	
B7	2279.60	1.39	1.16	0.66	0.54	0.31	0.9026	54.35	B7	0.561	0.561	
B8	2276.05	1.49	0.96	0.81	0.34	0.12	0.9677	85.35	B8	0.472	0.472	
B9	2264.15	1.36	1.2	1.15	0.4	0.12	0.9594	92.08	B9	0.404	0.404	
B10	2277.10	1.48	1.25	0.87	0.8	0.41	0.8979	49.18	B10	0.577	0.577	0.577
B11	2292.30	1.36	1.25	1.14	0.85	0.25	0.9159	66.81	B11	0.488	0.488	
B12	2289.10	1.43	0.83	0.7	0.26	0.15	0.9416	74.56	B12	0.479	0.479	
B13	2279.60	1.27	1.22	0.6	0.53	0.5	0.8662	49.76	B13	0.501	0.501	
B14	2287.10	1.05	0.71	0.35	0.24	0.21	0.9250	71.56	B14	0.474	0.474	
B15	2283.55	0.98	0.97	0.89	0.49	0.35	0.8941	64.96	B15	0.432	0.432	
B16	2290.20	1.4	1.25	0.74	0.66	0.5	0.8560	45.40	B16	0.515	0.515	
									MAX	0.577	0.577	
									MIN	0.404		
									STDEV	0.049138		
									AVG	0.495375		

TABLE III
TEST RESULTS OF THE 2ND CHARGE/DISCHARGE

The test result of the 2 nd experiment (3.6V 2200mAh)													
Battery no.	Rate capacity (mAh)	I ₁	I ₂	I ₃	I ₄	I ₅	NDC (%)	CT (min)	Battery no.	Fuzzy Output2	Fuzzy Output1	Pbest	Gbest
B1	2286.95	1.48	1.25	0.96	0.69	0.48	0.8619	43.91	B1	0.551	0.516	0.551	0.596
B2	2285.20	1.5	1.17	0.77	0.71	0.46	0.8720	44.98	B2	0.557	0.517	0.557	
B3	2266.90	1.47	1.41	1.02	0.59	0.26	0.9265	59.48	B3	0.572	0.417	0.572	
B4	2275.45	1.39	1.15	0.89	0.75	0.45	0.8811	47.23	B4	0.548	0.502	0.548	
B5	2278.25	1.19	1.11	0.86	0.83	0.52	0.8667	49.81	B5	0.501	0.515	0.515	
B6	2261.00	1.41	1.21	1.03	0.7	0.51	0.8699	44.68	B6	0.557	0.556	0.557	
B7	2277.10	1.44	1.18	0.87	0.71	0.41	0.9016	48.55	B7	0.596	0.561	0.596	
B8	2279.60	1.45	1.24	0.9	0.67	0.5	0.8604	43.46	B8	0.553	0.472	0.553	
B9	2264.15	1.5	1.21	0.66	0.56	0.39	0.8923	48.45	B9	0.557	0.404	0.557	
B10	2276.05	1.5	1.03	0.99	0.62	0.31	0.9141	55.63	B10	0.589	0.577	0.589	
B11	2292.30	1.5	1.29	1.09	0.67	0.35	0.8916	50.08	B11	0.548	0.488	0.548	
B12	2289.10	1.49	1.34	1.04	0.93	0.39	0.8798	51.08	B12	0.514	0.479	0.514	
B13	2279.60	1.39	1.21	1.18	0.81	0.45	0.8755	47.86	B13	0.532	0.501	0.532	
B14	2287.10	1.42	1.09	0.7	0.66	0.46	0.8649	46.05	B14	0.529	0.474	0.529	
B15	2283.55	1.35	1.06	0.95	0.7	0.54	0.8466	44.28	B15	0.497	0.432	0.497	
B16	2290.20	1.48	1.2	0.81	0.78	0.43	0.8666	47.43	B16	0.519	0.515	0.519	
									MAX	0.596	0.577	0.596	
									MIN	0.497	0.404		
									STDEV	0.028423	0.049138		
									AVG	0.545000	0.495375		

C. The Seventh Experiment

Table IV shows the test result of the seventh experiment. As in the prior experiments do, the obtained NDC and CT are inputted into the FDFE to evaluate the fitness value of each battery. The measured fitness values (fuzzy output 1 to output 7) of the 16 batteries from the first through the seventh experiments are illustrated in Table IV. Similarly, the data fields in Table IV marked with green indicate that these fitness values are local optimal for the particles under search, and the data field marked with blue indicate the G_{best} value. From Table

IV, the resulting STDEV in the seventh experiment is 0.007347; it has been less than 0.01. The difference of the AFV between the sixth and the seventh experiment is 0.003937; which is less than 0.005. Therefore, the specific termination conditions satisfied the default convergence criteria, and the PSO-based fuzzy-controlled searching process was stopped. The optimal solution found by the searching algorithm after 7 iterations is $[I_1, I_2, I_3, I_4, I_5] = [1.44, 1.18, 0.87, 0.71, 0.41]$ C. The corresponding waveforms of the battery current and voltage with temporal evolution is plotted in Fig. 9. From the measured waveform, the utilized programmable battery tester can accurately realize the proposed 5SCC charging algorithm.

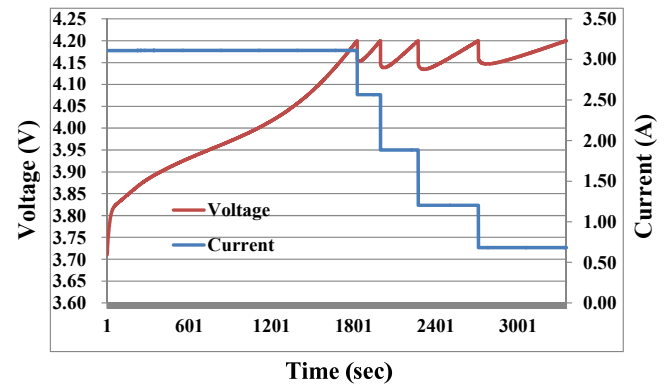


Fig. 9. Waveforms of the charging current and voltage.

D. Curves of Convergent Evolution

Shown in Fig. 10, 11, and 12 are the evolution trends of the NDC, CT and the calculated fitness value for each battery under test, respectively. From Fig. 12, all the evolution trends of the 16 particles gradually gather together and converge toward the leader particle. From Table IV, the AFV of each test increases steadily as the test number grows, which means the PSO algorithm is improving the obtained charging pattern during each experiment. It also implies that the particles are gradually gathering and following the trail of the leader particle. Moreover, the STDEV of the found solution decreases steadily. It demonstrates that the proposed searching algorithm can obtain a converged solution and feature fast convergence performance. Similarly, the behavior on the NDC and CT curves in Fig. 10 and 11 have the similar convergence characteristics.

E. Cycle Life Test

To verify the performance of the obtained final solution, battery cycling tests were conducted. Fig. 13 illustrates the remaining capacity (in %) versus cycle numbers for two new batteries from the same manufacturer (2200 mAh). In this study, two charging methods have been implemented to compare their performance. For conventional CC-CV method, a 1.44 C CC mode is utilized to charge the battery till the battery voltage reaches 4.2 V, and then a CV mode is applied until the charging current drops below 0.01C. For the proposed 5SCC method, the charging current in each stage is set to the obtained optimal value ($[I_1, I_2, I_3, I_4, I_5] = [1.44, 1.18, 0.87, 0.71, 0.41]$ C). The charger switches to the next stage when the battery voltage

TABLE IV TEST RESULTS OF THE SEVENTH CHARGE/DISCHARGE

The test result of the 7 th experiment(3.6V 2200mAh)																		
Battery no.	Rate capacity (mAh)	I ₁	I ₂	I ₃	I ₄	I ₅	NDC (%)	CT (min)	Battery no.	Fuzzy Output7	Fuzzy Output6	Fuzzy Output5	Fuzzy Output4	Fuzzy Output3	Fuzzy Output2	Fuzzy Output1	Pbest	Gbest
B1	2167.1	1.44	1.22	0.87	0.70	0.40	0.9013	51.25	B1	0.576	0.586	0.559	0.565	0.528	0.551	0.516	0.586	
B2	2169.9	1.43	1.18	0.86	0.71	0.40	0.9031	51.50	B2	0.581	0.566	0.558	0.572	0.531	0.557	0.517	0.581	
B3	2190.0	1.44	1.33	0.88	0.69	0.38	0.9084	52.11	B3	0.593	0.567	0.584	0.574	0.545	0.572	0.417	0.593	
B4	2180.8	1.42	1.17	0.87	0.69	0.41	0.9017	50.48	B4	0.583	0.597	0.557	0.582	0.561	0.548	0.502	0.597	
B5	2179.2	1.44	1.18	0.89	0.71	0.41	0.9007	50.01	B5	0.582	0.591	0.592	0.583	0.559	0.501	0.515	0.592	
B6	2124.8	1.45	1.19	0.79	0.71	0.40	0.9022	51.63	B6	0.577	0.566	0.543	0.569	0.546	0.557	0.556	0.577	
B7	2145.4	1.44	1.18	0.87	0.71	0.41	0.9046	50.21	B7	0.594	0.598	0.592	0.594	0.569	0.596	0.561	0.598	0.598
B8	2170.6	1.50	1.12	0.88	0.67	0.36	0.9098	54.05	B8	0.586	0.550	0.556	0.572	0.542	0.553	0.472	0.586	
B9	2158.2	1.43	1.25	0.78	0.69	0.39	0.9056	52.80	B9	0.581	0.572	0.593	0.575	0.553	0.557	0.404	0.593	
B10	2141.7	1.45	1.21	0.97	0.74	0.39	0.9056	52.76	B10	0.581	0.590	0.541	0.558	0.519	0.589	0.577	0.590	
B11	2114.3	1.47	1.17	0.93	0.72	0.42	0.8991	49.46	B11	0.580	0.592	0.593	0.559	0.546	0.548	0.488	0.593	
B12	2121.0	1.44	1.20	0.87	0.77	0.41	0.9008	51.55	B12	0.572	0.563	0.558	0.554	0.529	0.514	0.479	0.572	
B13	2164.9	1.44	1.18	0.87	0.71	0.41	0.9019	50.28	B13	0.585	0.586	0.579	0.570	0.537	0.532	0.501	0.586	
B14	2161.4	1.45	1.18	0.83	0.71	0.41	0.9004	50.80	B14	0.575	0.581	0.570	0.542	0.531	0.529	0.474	0.581	
B15	2154.5	1.45	1.19	0.87	0.71	0.41	0.8976	49.90	B15	0.573	0.553	0.550	0.543	0.517	0.497	0.432	0.573	
B16	2121.0	1.46	1.17	0.88	0.65	0.41	0.8959	49.00	B16	0.571	0.548	0.559	0.538	0.520	0.519	0.515	0.571	
									MAX	0.594	0.598	0.593	0.594	0.569	0.596	0.577	0.598	
									MIN	0.571	0.548	0.541	0.538	0.517	0.497	0.404		
									STDEV	0.006732	0.016883	0.018416	0.015739	0.015739	0.028423	0.049138		
									AVG	0.580625	0.575375	0.567750	0.565625	0.539563	0.545000	0.495375		

reaches 4.2 V. The 5SCC method stops when the voltage reaches 4.2 V at the 5th stage. It is noted that the charging current in the CC phase employed for the CC-CV method is the same as the current setting value of the first stage ($I_1=1.44$ C) to ensure fair comparison. From Fig. 13, the cycling number of the proposed algorithm when the capacity drop to 88% of the original capacity is 267; the cycling number of the conventional CC-CV method is 220 for the same capacity drop. That is, the proposed charge algorithm can provide 21% more cycle-life count than that of the conventional CC-CV method.

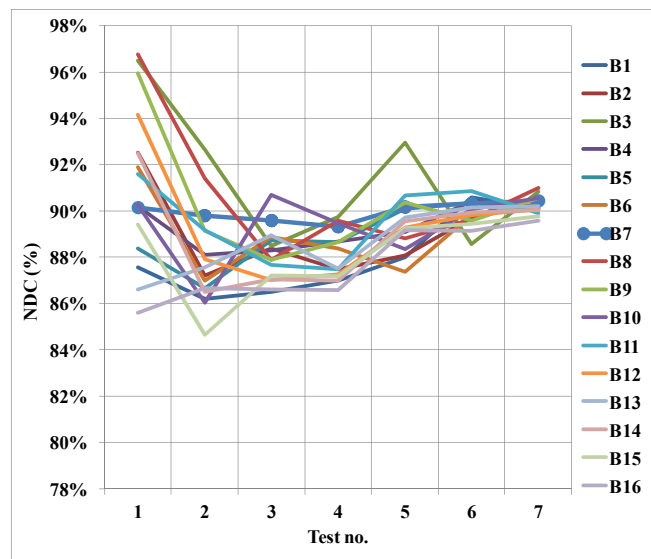


Fig. 10. Curves of the obtained NDC for each particle.

More records are shown to validate the salient performance of the proposed algorithm. Table V lists the charging time, charged capacity, discharged capacity, and charging efficiency obtained using the acquired optimal charging pattern and CC-CV method. The charging efficiency (CE) is definite by

$$CE = \frac{\text{Discharge capacity}}{\text{Capacity charged into battery during charging process}} \times 100\% \quad (11)$$

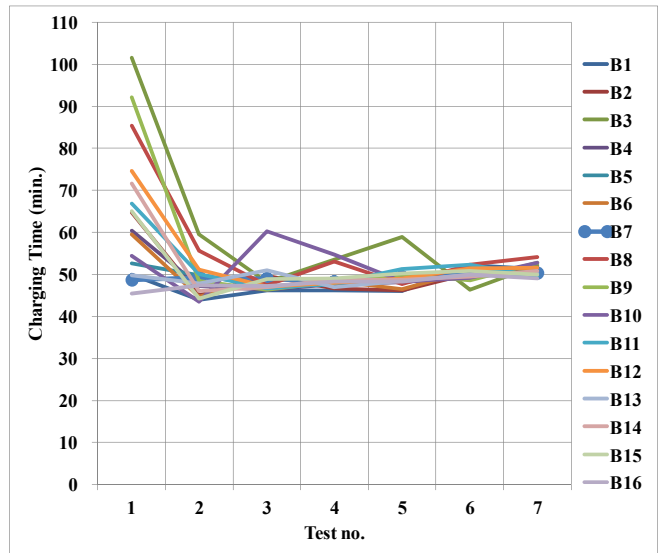


Fig. 11. Curves of the obtained CT for each particle.

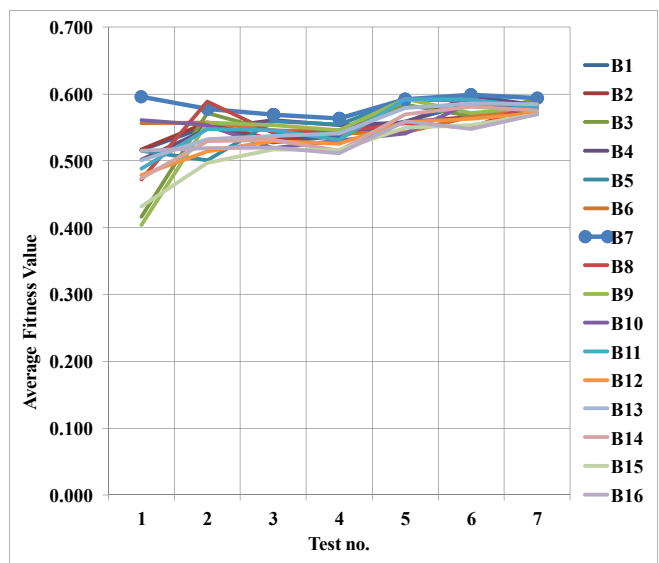


Fig. 12. Curves of the obtained average fitness value for each particle.

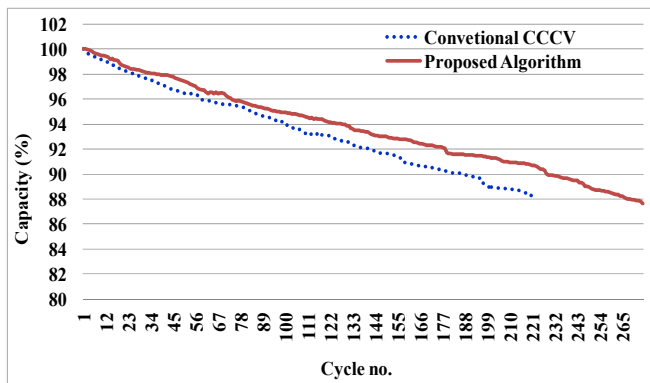


Fig. 13. Comparison of cycle life test between the proposed algorithm and the conventional CC-CV method.

TABLE V SALIENT PERFORMANCES FOR THE PROPOSED ALGORITHM

Proposed Algorithm (charge to 88% SOC)	Charge Time (min.)	51	Salient Performances	
	Charged Capacity (mAh)	1909.7	Charge Time Improvement (CTI, %)	$CTI = \frac{118 - 51}{118} \times 100\% = 56.8\%$
	Discharged Capacity (mAh)	1889.5		
	Charge Efficiency (%)	98.9 %	Charge Efficiency Improvement (CEI, %)	$CEI = 98.9\% - 98.5\% = 0.4\%$
	CC-CV Method (charge to 100% SOC)	Charge Time (min)		
Charged Capacity (mAh)		2145.6		
Discharged Capacity (mAh)		2114.3		
Charge Efficiency (%)		98.5%		

Although the charged capacity of the proposed charging pattern is lower than that of the CC-CV method, the charging time required has shown obvious improvement in 56.8%. Therefore, the obtained charging pattern is suitable for rapid charging application. The charging efficiency improvement in the proposed algorithm is 0.4%. The higher charge efficiency indicates fewer power losses, lower temperature increase produced during the charging process, and more cycling numbers.

VI. CONCLUSION AND DISCUSSION

This paper proposes a searching strategy, based on the PSO cooperating with the FDFE, to find the optimal multistage charging pattern for Li-ion batteries. The problem formulation, design consideration, implementation and experimental procedure, and dominant parameter setting mechanism have been described and explained in detail. After seven iterations, the searching procedure was stopped because the convergence criteria were met. The experimental results show that the obtained optimal charging pattern is capable of charging the batteries to over 88% capacity within 51 min. Compared with the conventional CC-CV method, the devised approach has a performance improvement of more than 56.8% charging time reduction, 21% more life cycles, and a 0.4% charging efficiency increase.

The experimental findings also show that the PSO-based fuzzy-controlled algorithm can be successfully incorporated into a commercially available battery tester to search for the optimal charging pattern. The advantages of the proposed method are: 1) Compared to other intelligent charging methods, the five-step CC charging algorithm can easily be implemented

and integrated into ICs. 2) The PSO-based fuzzy-controlled method is a favorable candidate for searching the optimal current setting values, as it is easy to implement and converges to the desired patterns quickly. 3) Compared to the conventional CC-CV method, the benefits of the obtained pattern include faster and safer charging and longer battery life-cycle. At the cost of slightly reduced discharged capacity, the devised charging approach is very suitable to the emerging applications in which the batteries are “sealed” inside the products to extend their lifespan. In addition, the parameter setting in the PSO model equation exist greater influence on the performance of the algorithm computation, which will be investigated in future work.

REFERENCES

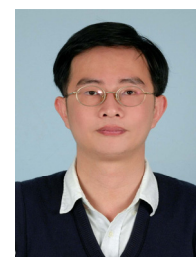
- [1] F. Hoffart, “Proper care extends Li-ion battery life,” *Power Electron. Technol. Mag.*, pp. 24–28, Apr. 2008.
- [2] S. Bhide and T. Shim, “Novel predictive electric Li-ion battery model incorporating thermal and rate factor effects,” *IEEE Trans. Veh. Technol.*, vol. 60, no. 3, pp. 819–829, Mar. 2011.
- [3] B. D. Valle, C. T. Wentz, and R. Sarpeshkar, “An area and power-efficient analog Li-ion battery charger circuit,” *IEEE Trans. Biomed. Circuits Syst.*, vol. 5, no. 2, pp. 131–137, Apr. 2011.
- [4] A. H. Ranjbar, A. Banaei, A. Khoobroo and B. Fahimi, “Online estimation of state of charge in Li-ion batteries using impulse response concept,” *IEEE Trans. Smart Grid*, vol. 3, no. 1, pp. 360–367, Mar. 2012.
- [5] H. T. Lin, T. J. Liang, and S. M. Chen, “Estimation of battery state of health using probabilistic neural network,” *IEEE Trans. Ind. Inf.*, vol. 9, no. 2, pp. 679–685, May 2013.
- [6] L. Lam and P. Bauer, “Practical capacity fading model for Li-ion battery cells in electric vehicles,” *IEEE Trans. Power Electron.*, vol. 28 no. 12, pp. 5910–5918, Dec. 2013.
- [7] M. E. Orchard, P. Hevia-Koch, Z. Bin, and T. Liang, “Risk measures for particle-filtering-based state-of-charge prognosis in lithium-ion batteries,” *IEEE Trans. Ind. Electron.*, vol. 60, no. 11, pp. 5260–5269, Nov. 2013.
- [8] M. Chen and G. A. Rincon-Mora, “Accurate, compact, and power-efficient Li-ion battery charger circuit,” *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 53, no. 11, pp. 1180–1184, Nov. 2006.
- [9] R. Pagano, M. Baker, and R. E. Radke, “A 0.18-μm monolithic Li-ion battery charger for wireless devices based on partial current sensing and adaptive reference voltage,” *IEEE J. Solid-State Circuits*, vol. 47, no. 6, pp. 1355–1368, Jun. 2012.
- [10] S. H. Yang, J. W. Liu, and C. C. Wang, “A single-chip 60-V bulk charger for series Li-ion batteries with smooth charge-mode transition,” *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 59, no. 7, pp. 1588–1597, Jul. 2012.
- [11] H. Rahimi-Eichi, U. Ojha, F. Baronti, and M. Chow, “Battery management system: An overview of its application in the smart grid and electric vehicles,” *IEEE Trans. Ind. Electron. Mag.*, vol. 7, no. 2, pp. 4–16, Jun. 2013.

- [12] G. Pellegrino E. Armando, and P. Guglielmi, "An integral battery charger with power factor correction for electric scooter," *IEEE Trans. Power Electron.*, vol. 25 no. 3, pp. 751-759, Mar. 2010.
- [13] A. Hoke, A. Brissette, K. Smith, A. Pratt, and D. Maksimovic "Accounting for Lithium-ion battery degradation in electric vehicle charging optimization," *IEEE J. Emerging Sel. Topics Power Electron.*, vol. 2 no. 3, pp. 751-759, Sept. 2014.
- [14] E. Inoa and J. Wang, "PHEV charging strategies for maximized energy saving," *IEEE Trans. Veh. Technol.*, vol. 60, no. 7, pp. 2978-2986, Sept. 2011.
- [15] J. Jiang, Q. Liu, C. Zhang, and W. Zhangnd, "Evaluation of acceptable charging current of power Li-ion batteries based on polarization characteristics," *IEEE Trans. Ind. Electron.*, vol. 61, no. 12, pp. 6844-6851, Dec. 2014.
- [16] T. C. Huang; R. H. Peng, T. W. Tsai, K. H. Chen, and C. L. Wey, "Fast charging and high efficiency switching-based charger with continuous built-in resistance detection and automatic energy deliver control for portable electronics," *IEEE J. Solid-State Circuit*, vol. 49, no. 7, pp. 1580-1594, Jul. 2014.
- [17] A. Kuperman, U. Levy, J. Goren, A. Zafransky, and A. Savernin, "Battery charger for electric vehicle traction battery switch station," *IEEE Trans. Ind. Electron.*, vol. 60, no. 12, pp. 5391-5399, Dec. 2013.
- [18] S. Chae, Y. Song, S. Park, and H. Jeong, "Digital current sharing method for parallel interleaved DC-DC converters using input ripple voltage," *IEEE Trans. Ind. Inf.*, vol. 8, no. 3, pp. 536-544, Aug. 2012.
- [19] J. Jiang, C. Zhang, J. Wen, W. Zhang, and S. M. Sharkh, "An optimal charging method for Li-ion batteries using a fuzzy-control approach based on polarization properties," *IEEE Trans. Veh. Technol.*, vol. 62, no. 7, pp. 3000-3009, Sept. 2013.
- [20] H. Rahimi-Eichi, F. Baronti, and M. Y. Chow, "Online adaptive parameter identification and state-of-charge coestimation for lithium-polymer battery cells," *IEEE Trans. Ind. Electron.*, vol. 61, no. 4, pp. 2053-2061, Apr. 2014.
- [21] Y. C. Hsieh and C. S. Huang, "Li-ion battery charger based on digitally controlled phase-shifted full-bridge converter," *IET Power Electron.*, vol. 4, no. 2, pp. 242-247, Feb. 2011.
- [22] C. H. Lin, C. Y. Hsieh, and K. H. Chen, "A Li-ion battery charger with smooth control circuit and built-in resistance compensator for achieving stable and fast charging," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 57, no. 2, pp. 705-712, Feb. 2010.
- [23] P. Ramadass, B. Haran, P. M. Gomadam, R. White, and B. N. Popov, "Development of first principles capacity fade model for Li-ion cells," *J. the Electrochemical Society*, vol. 151, no. 2, pp. A196-A203, 2004.
- [24] L. R. Chen, C. S. Liu, J. J. Chen, "Improving phase-locked battery charger speed by using resistance-compensated technique," *IEEE Trans. Ind. Electron.*, vol. 56, no. 4, pp. 1205-1211, Apr. 2009.
- [25] L. R. Chen, "Design of duty-varied voltage pulse charger for improving Li-ion battery-charging response," *IEEE Trans. Ind. Electron.*, vol. 56, no. 2, pp. 480-487, Feb. 2009.
- [26] L. R. Chen, J. J. Chen, N. Y. Chu, and G. Y. Han, "Current-pumped battery charger," *IEEE Trans. Ind. Electron.*, vol. 55, no. 6, pp. 2482-2488, Jun. 2008.
- [27] L. R. Chen, "A design of an optimal battery pulse charge system by frequency-varied technique," *IEEE Trans. Ind. Electron.*, vol. 54, no. 1, pp. 398-405, Feb. 2007.
- [28] T. Ikeya, N. Sawada, S. Takagi, J. Murakami, and K. Kobayashi et al., "Multi-step constant-current charging method for electric vehicle, valve-regulated, lead/acid batteries during night time for load-leveling," *J. Power Sources*, vol. 75, no. 1, pp. 101-107, Sept. 1998.
- [29] T. Ikeyaa, N. Sawadab, J. I. Murakamic, K. Kobayashi, M. Hattori, N. Murotani, S. Ujiie, K. Kajiyama, H. Nasu, H. Narisoko, Y. Tomaki, K. Adachi, Y. Mita, and K. Ishihara, "Multi-step constant-current charging method for an electric vehicle nickel/metal hydride battery with high energy efficiency and long cycle life," *J. Power Sources*, vol. 105, no. 1, pp. 6-12, 2002.
- [30] P. C. Lin, Y. H. Liu, Y. H. Liu, J. K. Chen, and C. H. Chen, "A fully digital rapid charger for electric scooters," in *Proc. Eighteenth Symp. Electrical Vehicles*, 2001, Session D7A, pp. 1-13.
- [31] J. W. Huang, Y. H. Liu, S. C. Wang, and Z. Z. Yang, "Fuzzy-control-based five-step Li-ion battery charger," in *Proc. IEEE Int. Conf. PEDS*, Nov. 2009, pp. 1547-1551.
- [32] L. R. Dung and J. H. Yen, "ILP-based algorithm for Lithium-ion battery charging profile," in *Proc. IEEE Int. Symp. Ind. Electron.*, 2010, pp. 2286-2291.
- [33] Z. Ullah, B. Burford, and S. Dillip "Proper care extends Li-ion battery life," *IEEE Aerospace and Electron. Syst. Mag.*, vol. 11, no. 6, pp. 26-34, Jun. 1996.
- [34] F. J. Lin, M. S. Huang, P. Y. Yeh, H. C. Tsai, and C. H. Kuan, "DSP-based probabilistic fuzzy neural network control for Li-ion battery charger," *IEEE Trans. Power Electron.*, vol. 27 no. 8, pp. 3782-3794, Aug. 2012.
- [35] M. Charkhgard and M. Farrokhi, "State-of-charge estimation for lithium-ion batteries using neural networks and EKF," *IEEE Trans. Ind. Electron.*, vol. 57, no. 12, pp. 4178-4187, Dec. 2010.
- [36] K. Thirugnanam, J.T.P. Ezhil Reena, M. Singh, and P. Kumar, "Mathematical modeling of Li-ion battery using genetic algorithm approach for V2G applications," *IEEE Trans. Energy Convers.*, vol. 29, no. 2, pp. 332-343, Jun. 2014.
- [37] H. Surmann, "Genetic optimization of a fuzzy system for charging batteries," *IEEE Trans. Ind. Electron.*, vol. 43, no. 5, pp. 541-548, Oct. 1996.
- [38] S. Bashash, S. J. Moura, J. C. Forman, H. K. Fathy, "Plug-in hybrid electric vehicle charge pattern optimization for energy cost and battery longevity," *J. Power Sources*, vol. 196, no. 1, pp. 541-549, 2011.
- [39] Y. H. Liu, J. H. Teng, and Y. C. Lin, "Search for an optimal rapid charging pattern for lithium-ion batteries using ant colony system algorithm," *IEEE Trans. Ind. Electron.*, vol. 52, no. 5, pp. 1328-1336, Oct. 2005.
- [40] Y. H. Liu and Y. F. Luo, "Search for an optimal rapid-charging pattern for Li-ion batteries using the Taguchi approach," *IEEE Trans. Ind. Electron.*, vol. 57, no. 12, pp. 3963-3971, Dec. 2010.
- [41] Y. H. Liu, C. H. Hsieh, and Y. F. Luo, "Search for an optimal five-step charging pattern for Li-ion batteries using consecutive orthogonal arrays," *IEEE Trans. Energy Convers.*, vol. 26, no. 2, pp. 654-661, Jun. 2011.
- [42] J. Kennedy and R. C. Eberhart, "Particle swarm optimization," in *Proc. IEEE Int. Conf. Neural Networks*, 1995, pp. 1942-1948.
- [43] R. C. Eberhart and Y. H. Shi, "Particle swarm optimization: developments, applications and resources," in *Proc. IEEE Int. Cong. Evolutionary Computation*, 2001, pp. 81-86.



Shun-Chung Wang received the Ph.D. degrees in electrical engineering from National Taiwan University Taipei, Taiwan, in 1995.

He is currently with the Department of Electrical Engineering, Lunghwa University of Science and Technology, Taoyuan, Taiwan. His current research interests include power electronics and their applications.



Yi-Hwa Liu (M'01) received the Ph.D. degree in electrical engineering from National Taiwan University, Taipei, Taiwan, in 1998.

He joined the Department of Electrical Engineering, Chang-Gung University, Taoyuan, Taiwan, in 2003. He is currently with the Department of Electrical Engineering, National Taiwan University of Science and Technology, Taipei. His current research interests include power electronics and battery management.